



MECH MAINT Department  
SIMHADRI

**Note No :1**



**Ref:**SMTTP/MECH MAINT/2023-24/LMI/454565

**Date:**05/09/2023(11:32:33)

**Subject:**Approval for Revision of LMI

Wrt the revised TC document no: COS-ISO-00-OGN/OPS/PERF/002 Rev 02 Aug. 2023, The Location Management Instruction (LMI) of Performance testing of condenser air extraction plant i.e LMI/SMSTPS/EEMG-07/OGN/OPS/PERF/002 revision is done.

Approving authority for LMI is HOP.

Submitted for kind approval.

**Submitted by:**

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**Note No:2**

LMI/SMSTPS/EEMG-07/OGN/OPS/PERF/002 rev 03 Aug,23 may pls. be approved.

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**Note No:3**

Please review

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**Note No:4**

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**Note No:5**

LMI reviewed, no change is suggested.

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**On:**06/09/2023(12:19:33) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/454565

**Note No:6**

Forwarded for approval please.

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**Note No:7**

May please approve

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**Note No:8**

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**On:**07/09/2023(09:28:21) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/454565

***NTPC Ltd***

**LOCATION MANAGEMENT INSTRUCTION**

**Performance Testing of Condenser Air  
Extraction plant**

**LMI/SMSTPS/EEMG-07/OGN/OPS/PERF/002**

**Rev: 03 August 2023**

Approved for Implementation by: **HOP (SMSTPS)**

Enquiries to: **HOD (EEMG)**

|               |                                                               |
|---------------|---------------------------------------------------------------|
| <b>TITLE</b>  | <b>Performance Testing of Condenser Air Extraction Plant</b>  |
| <b>REF NO</b> | <b>LMI/SMSTPS/EEMG-07/OGN/OPS/PERF/002 Rev:03 August 2023</b> |

## LMI APPROVAL DOCUMENT

|   |                     |                                                         |
|---|---------------------|---------------------------------------------------------|
| 1 | LMI NAME            | Performance Testing of Condenser Air Extraction Plant   |
| 2 | LMI No.             | LMI/SMSTPS/EEMG-07/OGN/OPS/PERF/002 Rev: 03             |
| 3 | Reference Document  | OGN/OPS/PERF/002/ Rev:02 August:2023                    |
| 4 | Superseded Document | LMI/SMSTPS/ OGN/OPS/PERF/002/ Rev:02 Apr 2022           |
| 5 | Prepared By         | Name: Puspita Behera<br>Designation: Sr. Manager (EEMG) |
| 6 | Checked By          | Name: Ajay Gupta<br>Designation: DGM (EEMG)             |
| 7 | Reviewed By         | Name: Uma Shanker Verma<br>Designation: AGM (EEMG)      |
| 8 | Approved By         | Name: Sanjay Kumar Sinha<br>Designation: GGM (Simhadri) |

|               |                                                               |
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| <b>TITLE</b>  | <b>Performance Testing of Condenser Air Extraction Plant</b>  |
| <b>REF NO</b> | <b>LMI/SMSTPS/EEMG-07/OGN/OPS/PERF/002 Rev:03 August 2023</b> |

## **1.0 INTRODUCTION:**

The condenser air extraction plant is provided to rapidly evacuate air and any other non-condensable gases during hogging & holding operation, from the condenser steam space and associated pipework on start up, and to maintain operational vacuum conditions in the condenser with the machine on load.

The plant design at NTPC Simhadri super thermal power station is having vacuum pumps of liquid ring type.

The performance of the air extraction system can be affected by secondary cooler fouling, internal plant leakages, mechanical defects of pumps and protection devices.

In addition, the performance of this type of plant varies through out the year (as does that of condenser) with the change is CW inlet temperature from which sub system cooling water is drawn. This will also affect the suction conditions, varying the water vapor/air ratio to be handled by the plant.

Poor condenser vacuum, caused by air blanketing, may be due to leakage well in the excess of air extraction plant capacity or a direct shortfall in the performance of the air extraction plant. This results from the general difficulty in identifying condenser air leaks, a simple test to determine the air extraction plant performance is therefore essential to identify the primary cause of any problem.

## **2.0 SUPERSEDED DOCUMENTS:**

LMI/OGN/OPS/PERF/002 Rev:02, Apr 2022

## **3.0 RECORDS OF REVISION:**

The revision is done due to the TC document revision.

## **4.0 SCOPE:**

The scope of testing extends to vacuum pumps used for air extraction of Simhadri station.

## **5.0 PROCEDURES:**

The procedure for measurement of the performance of power station condenser air extraction plant in respect of its air handling capacity is as follows:

Plant testing conditions:

- i) Unit should be at full load in normal stable condition.
- ii) Vacuum pump seal water flow, temperature & pressure should be normal range.



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## 5.1 SINGLE VACUUM PUMP OPERATION:

5.1.1 Record all the parameters as per annexure - I when single pump is running

- i) changeover of running pump with stand by pump and record all parameters as per annexure-I.
- ii) compare condenser back pressure after correction for load and CW inlet temperature, when two pumps are running separately.

Remarks: corrected condenser back pressure should be same for both vacuum pumps. Otherwise, higher back pressure with any pump in service indicates poor performance of the pump.

## 5.2 TWO VACCUM PUMPS IN PARALLEL OPERATION:

5.2.1 Record all parameters as per annexure-I when any one pump is running

- i) Start stand by vacuum pump and record all parameters as per annexure-I
- ii) Observe deviation in condenser back pressure after correcting for load and CW inlet Temp.

Remarks: Deviation in condenser back pressure more than three mmHg indicates air ingress in condenser is more than the capacity of a single pump.

## 6.0 CALCULATION:

- i) Derive saturation temperature corresponding vacuum pump inlet pressure. Vacuum pump operating water temperature should less than saturation temperature corresponding vacuum pump inlet pressure. .... **EEMG**
- ii) Difference between sealing water & cooling water temperature should be approximately 2.5 degree C max. If the temperature is more, then heat exchanger effectiveness is to be checked. .... **EEMG/TMD**
- iii) Trending of rotameter reading to be done for early intervention of air ingress. .... **EEMG**
- iv) Helium leak test is to be conducted for identification and rectification of air ingress points. After attending the defects condenser back pressure improvements to be recorded. .... **TMD**

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## 7.0 REVIEW:

The head of project is responsible for reviewing this LMI on two yearly basis or as or necessary.

Annexure-I

### VACUUM (LIQUID RING) PUMP LOG SHEET (ROUTINE CHECK LIST)

| Sl no | Parameter                                                                          | Unit        | Running parameter |
|-------|------------------------------------------------------------------------------------|-------------|-------------------|
| 1     | Unit Load                                                                          | MW          |                   |
| 2     | Condenser Vacuum                                                                   | mmHg        |                   |
| 3     | Vacuum pump sealing water temp.                                                    | deg. C      |                   |
| 4     | Cooling water inlet pressure & temperature of Heat Exchanger                       | Ksc/deg. C  |                   |
| 5     | Cooling water (Working fluid) outlet pressure & temperature temp of Heat Exchanger | Ksc/deg. C  |                   |
| 6     | Strainer (sealing water) DP                                                        |             |                   |
| 7     | Separator level                                                                    |             |                   |
| 8     | Separator Air flow (Rotameter reading)                                             | Kg/hr       |                   |
| 9     | Recirculation pump flow                                                            | m3/h        |                   |
| 10    | Vacuum Pump Motor current                                                          | A           |                   |
| 11    | Direction of rotation of motor viewed from motor NDE side.                         |             |                   |
| 12    | Instrument air pressure                                                            | Ksc.        |                   |
| 13    | Feed water flow valve of separator tank status.                                    | Auto/manual |                   |
| 14    | No. Vacuum pumps running                                                           |             |                   |

|               |                                                               |
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## Annexure-II

**DESIGN VALUE AND RUNNING PARAMETER**

| <b>S. No.</b> | <b>Parameter</b>                                    | <b>Unit</b> | <b>Design value</b> | <b>Running parameter</b> |
|---------------|-----------------------------------------------------|-------------|---------------------|--------------------------|
| 1.            | Unit Load                                           | MW          |                     |                          |
| 2.            | Air flow rate                                       | Kg/Hr       |                     |                          |
| 3.            | Condenser absolute pressure                         | Ksc         |                     |                          |
| 4.            | Air extraction plant suction absolute pressure      | Ks<br>c     |                     |                          |
| 5.            | Intermediate air extraction plant absolute pressure | Ks<br>c     |                     |                          |
| 6.            | Air extraction plant suction temperature            | Deg C       |                     |                          |
| 7.            | Condenser CW inlet temperature/outlet temperature   | Deg C       |                     |                          |
| 8.            | Ambient air temperature                             | Deg C       |                     |                          |
| 9.            | Ambient air pressure                                | Ks<br>c     |                     |                          |
| 10.           | Cooling water/spray water flow rates                | Kg/s,LPM    |                     |                          |
| 11            | Auxiliary cooler inlet and outlet temperatures      | Deg C       |                     |                          |